
IM0802 – Responsible Artificial Intelligence

Assignment 2: “Paper Title Goes Here”

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Abstract

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1 Introduction 350

Israel, which allocates 8.8% of its GDP to defense [1], invests heavily in the development of autonomous weapon systems (AWS), with their Prime Minister announcing a \$100 billion investment in homegrown arms production last December [2]. Israel’s most well-known AWS is the Iron Dome air defense system, which intercepts incoming projectiles in real time. Building on this foundation in artificial intelligence (AI)-assisted defense technology, Israel has expanded into offensive targeting applications such as “Habsora”.

Habsora ("The Gospel") is an Israeli Defense Forces (IDF) AI-targeting system developed primarily by Unit 8200 within the Military Intelligence Directorate. First deployed during the May 2021 Operation "Guardian of the Walls", it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. The system is not a single model, but an ensemble of multiple specialised algorithms fused together [4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which in prior years required approximately twelve months of human analysts [3, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations and individuals as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis [6]. Military commands nominally authorise each recommendation, yet their review window documented for each target was approximately twenty seconds [3].

This creates a stakeholder dynamic with both ethical and legal significance. It’s output influencing both IDF commands, who operate under high cognitive load and time pressure; Palestinian civilians in the target environment bearing the lethal consequences and the broader international community which depends on International Humanitarian Law (IHL) compliance for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control, which both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines the second condition, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdiesen et al. [8] in their Comprehensive Human Oversight Framework (CHOF).

2 Ethical Issue Identification 750

2.1 Governance

Article 36 of Protocol I to the Geneva Conventions requires countries to perform legal reviews of new weapons systems to determine whether their use would be prohibited under international law [9]. Israel is one of the few countries that have not ratified Protocol I [10]. This means it has no legal obligation to review Habsora's compliance with IHL prior to deployment. UNESCO maintains that delegating life-or-death decisions to an AI system is itself forbidden [11]. Yet, without ratification, no mechanism exists to enforce this in the case of Habsora **TODO EVIDENCE NEEDED**.

Once deployed, establishing accountability becomes difficult. Under IHL, commanders are legally responsible for how weapons are used, even when autonomous functions are involved [12]. They are judged on what they knew at the time of the decision [13]. In practice, responsibility is spread across developers, the algorithm, the receiving IDF officer, and the approving commander [11]. The system has a documented 10% error tolerance in targeting recommendations. This illustrates the problem: this margin makes civilian casualties inevitable. In the midst of war, real-time monitoring is not easy to pull off, which compounds the issue.

After deployment, we find that review processes are largely absent. The system's opacity makes it difficult to determine whether casualties resulted from an AI recommendation or a human override. Palestinian civilians, who the most directly affected stakeholders, were never consulted. No structured ethical assessment, such as 'value sensitive design' (VSD), was conducted.

2.2 Socio-technical

Palestinian civilians, the stakeholders most directly affected by Habsora, were not involved in its design and their values were not elicited to ground the system's targeting logic **TODO EVIDENCE NEEDED**. They remain non-consenting parties who face the consequences of the system's use without recourse. There is a large asymmetry in consequences between those operating the system and those subjected to it.

For IDF commanders, the conditions necessary for genuine human oversight are structurally not present. Steen et al. [7] argue that meaningful human control requires both sufficient information and meaningful freedom of action. Habsora undermines both: commanders receive recommendations without transparency into the underlying reasoning, and the short decision windows of the system leave almost no room for critical evaluation. Miller [14] shows that from social science, without understanding what went into a decision, the explainee cannot meaningfully interrogate it. Combined with the large amount of targets the system generates, this reduces what would normally require extended deliberation and proportionality assessments to a button press. Former IDF commanders have described this process as 'rubber-stamping' [15]. This confirms the presence of automation bias, which is when operators rely too much on AI outputs under time pressure [16].

The consequence is that IDF commanders formally bear moral and legal responsibility for strikes they cannot meaningfully evaluate, while the system's opacity diffuses accountability in practice. Targeting decisions that once demanded careful human judgement are now effectively delegated to an algorithm. This prevents the human control which the IHL demands, and that Steen et al. [7] identify as a precondition for attributing responsibility.

References

- [1] Stockholm International Peace Research Institute. Trends in world military expenditure, 2024. SIPRI, April 2025. URL https://www.sipri.org/sites/default/files/2025-04/2504_fs_millex_2024.pdf. SIPRI Fact Sheet. Accessed: 07 April 2026.
- [2] Times of Israel Staff. Israel investing more than \$100 billion into independent arms industry, Netanyahu reveals. *The Times of Israel*, December 2025. URL <https://www.timesofisrael.com/israel-investing-more-than-100-billion-into-independent-arms-industry-netanyahu-reveals/>. Accessed: 12 April 2026.
- [3] Yuval Abraham. "lavender": the ai machine directing israel's bombing spree in gaza, Apr 2024. URL <https://www.972mag.com/lavender-ai-israeli-army-gaza/>.

- [4] Elizabeth Dwoskin. Israel built an “ai factory” for war. it unleashed it in gaza., Dec 2024. URL <https://www.washingtonpost.com/technology/2024/12/29/ai-israel-war-gaza-idf/>.
- [5] Ynet. “man replaced by machine”: Is the use of ai undermining the idf’s intelligence capabilities?, Dec 2024. URL <https://www.ynetnews.com/article/sjyhbn1ikx>.
- [6] Michael N. Schmitt and Jeffrey S. Thurnher. "out of the loop": Autonomous weapon systems and the law of armed conflict. *Harvard National Security Journal*, 4(2):231–281, 2013. URL <https://centaur.reading.ac.uk/89863/>.
- [7] Marc Steen, Jurriaan Diggelen, Tjerk Timan, and Nanda Stap. Meaningful human control of drones: exploring human–machine teaming, informed by four different ethical perspectives. *AI and Ethics*, 3, 05 2022. doi: 10.1007/s43681-022-00168-2.
- [8] Ilse Verdiesen, Andrea Aler Tubella, and Virginia Dignum. Integrating comprehensive human oversight in drone deployment: A conceptual framework applied to the case of military surveillance drones. *Information*, 12(9), 2021. ISSN 2078-2489. doi: 10.3390/info12090385. URL <https://www.mdpi.com/2078-2489/12/9/385>.
- [9] International Committee of the Red Cross. Protocol additional to the geneva conventions of 12 august 1949, and relating to the protection of victims of international armed conflicts (protocol i), 8 june 1977, article 36 - new weapons. <https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/article-36>, 1977. ICRC Database, Treaties, States Parties and Commentaries. Last accessed: 22 April 2026.
- [10] International Committee of the Red Cross. State parties to Additional Protocol I of 1977. IHL Treaties Database, 2024. URL <https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/state-parties>.
- [11] Sherry Wasilow and Joelle B. Thorpe. Artificial intelligence, robotics, ethics, and the military: A canadian perspective. *AI Magazine*, 40(1):37–48, 2019. doi: <https://doi.org/10.1609/aimag.v40i1.2848>. URL <https://onlinelibrary.wiley.com/doi/abs/10.1609/aimag.v40i1.2848>.
- [12] International Committee of the Red Cross. Protocol I additional to the Geneva Conventions of 12 august 1949, article 86: Failure to act. IHL Treaties Database, 1977. URL <https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/article-86>.
- [13] Jean-Marie Henckaerts and Louise Doswald-Beck. *Customary International Humanitarian Law*, volume 1. Cambridge University Press, Cambridge, 2005.
- [14] Tim Miller. Explanation in artificial intelligence: Insights from the social sciences, 2018. URL <https://arxiv.org/abs/1706.07269>.
- [15] Atay Kozlovski. When algorithms decide who is a target: Idf’s use of ai in gaza. *Tech Policy Press*, May 2024. URL <https://www.techpolicy.press/when-algorithms-decide-who-is-a-target-idfs-use-of-ai-in-gaza/>.
- [16] Raja Parasuraman and Dietrich H. Manzey. Complacency and bias in human use of automation: An attentional integration. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 52(3):381–410, 2010. doi: 10.1177/0018720810376055.